

Scientific Referee Assessment

*The Face-Adjacency Graph of the Truncated Octahedron:
Laplacian Spectrum, Symmetry Decomposition, an Inter-Orbit
Operator, and the Magnetic Laplacian*

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Recommendation	Accept with minor editorial/release revisions, subject to specialist venue fit.
Mathematical validity	No critical or major mathematical error confirmed within the assessed scope; this is not a formal correctness certificate.
Assessment score	9.2/10 (rounded assessment score, not a probability of correctness).
Identified remaining task	One substantive submission-stage archival/reproducibility action; two optional clarity improvements.
Comparison with Draft 6	Core mathematical results unchanged; proof-status, attribution, figure scope, and verification-history statements are clearer.

Executive assessment

Draft 7 is a focused, well-supported mathematical note about a 14-vertex, nonregular face-adjacency graph. Its strongest features are the explanation of the ordinary spectrum by representation blocks, the exact quarter-flux Gram reduction and magnetic rotation action, and the third spectral-moment proof that there are exactly thirteen distinct uniform-flux spectra. The comparison with the five Fedorov parallelohedra is useful when kept descriptive rather than elevated to a physical selection principle.

The principal findings are supported by hand arguments in the paper and by independent finite reconstructions: the graph’s incidence pattern and characteristic polynomials, the restricted K identities, the $q = 6$ Gram identity, the $q = 12$ signless Laplacian, the magnetic moments, the spanning-tree count, and the 24-element rotation action. The displayed mathematics was the subject of these checks; the precise `face-graph-note-v7` software release was *not* independently installed and executed in this round. This limitation affects the release-audit claim, not the verified spectral calculations.

Compared with Draft 6, the manuscript has addressed the previous concern about lengthy review-history detail and now makes an appropriate distinction between human-readable mathematical proof, finite exact checking, and floating-point corroboration. It appropriately disclaims proof-assistant certification. A permanent archive/identifier is still promised for submission rather than cited as already deposited. This is the sole remaining actionable condition in this assessment.

Contents

Executive assessment	1
1 Scientific contribution and scope	2

2	Category-level assessment	2
3	Detailed metric assessment	3
4	Independent mathematical verification	4
4.1	Graph geometry and characteristic polynomials (pp. 2–4)	4
4.2	Eigenspace support and the K operator (pp. 4–5)	4
4.3	Flux classification and third moment (pp. 5–7)	5
4.4	Quarter and half flux (pp. 7–9)	5
4.5	Fourth moment, tree count and Fedorov table (pp. 6, 9–13)	6
5	Detailed findings and correction status	6
6	Effect of findings on dependent results	7
7	Author and editor recommendations	7
8	Overall recommendation	7

1 Scientific contribution and scope

The underlying face graph is not itself claimed as newly discovered: it is the tetrakis-hexahedral graph. Rather, the contribution lies in assembling several exact, mutually consistent properties of its Laplacian and two nontrivial magnetic sectors, with explicit symmetric reductions. The manuscript correctly separates three different achievements:

- an *exact zero-flux eigenspace decomposition*, including support on square and hexagon vertices;
- *exact closed forms for two special fluxes*, $q = 6$ and $q = 12$, together with a magnetic representation at $q = 6$;
- an *exact classification of spectral equality across all integer charges*, which does not imply exact closed-form eigenvalues at every charge.

The physical motivation from foam-cell geometry is explicitly outside this mathematical note. That restraint is scientifically important: the five-cell spectral comparison is not a variational or dynamical argument selecting the truncated octahedron as the physical microscopic cell.

2 Category-level assessment

The paper is assessed as pure mathematics with exact finite computations. Fourteen core criteria are scored, with gauge and algebraic specializations active. The table is an editorial assessment of the supplied version; a discovered mathematical error would override a favourable aggregate.

Category	Weight	Score / 10	Contribution
Mathematical correctness (M1–M4)	45.0%	9.295	4.183
Completeness (M5–M7)	22.5%	9.380	2.111
Clarity (M8–M10)	18.0%	9.230	1.661
Significance (M11–M14)	4.5%	6.220	0.280
Gauge structure (M15)	7.0%	9.300	0.651
Exact algebraic structure (M18)	3.0%	9.500	0.285
Total	100%		9.171 / 10

The category averages retain the existing within-category proportions (10:12:10:8; 10:8:7; 7:7:6; 5:3:4:3). The reported overall score is rounded to 9.2. Narrow generality and uncertain priority limit the significance category; they do not imply an incorrect mathematical claim.

3 Detailed metric assessment

Metric	Criterion	Score	Manuscript-specific justification
M1	Theorem precision	9.3	The graph, symmetry group, finite sectors, and restricted intertwiner are specified. Prop. 6.2 distinguishes spectral equivalence from explicit eigenvalues.
M2	Proof completeness	9.2	Orbit decomposition, gauge classification, Gram factorization and third-moment argument give complete main proof chains. Standard character identifications remain compact.
M3	Calculation accuracy	9.4	Exact reconstruction reproduces ordinary, signless and quarter-flux polynomials, the fourth moment and the spanning-tree count.
M4	Logical consistency	9.3	The two K^2 values concern separate invariant sectors; $E_7 = A_{1g} \oplus T_{2g}$ is distinguished from an irreducible eigenspace.
M5	Definitions	9.2	Edge graph, A , D , Δ , orbit projections, eigen-support weights, gauge transformations and admissible flux sectors are defined.
M6	Assumptions and evidence	9.5	Spherical embedding and uniform-flux integer condition are explicit; the meaning of proof/exact/numeric and absence of Lean validation are now particularly clear (p. 11).
M7	Scope	9.5	The numerical spectral sweep is not confused with analytic eigenvalue branches or with proof of arbitrary-charge factorizations. Physical cell-selection claims are expressly excluded.
M8	Notation	9.2	Orbit and O_h labels are mostly clear; $A_{hs} = A_{sh}^T$ is provided. Several dense displays could be introduced verbally.
M9	Accessibility	9.2	Raw and orthonormal 2×2 blocks make the argument accessible to a graph-spectrum reader; the five-step roadmap in §6.1 is effective.
M10	Organization	9.3	The introduction anticipates the later sections; the clarified acknowledgements now avoid repeating a procedural history at length.
M11	Originality	6.0	Explicit example-specific magnetic reductions plausibly contribute useful calculations; exhaustive historical priority has not been established.
M12	Generality	4.0	Exact special-flux forms apply to a single 14-vertex graph; no general new theorem for arbitrary polyhedral tilings is claimed.
M13	Scholarship	8.7	Relevant spectral, magnetic and polyhedral antecedents are cited and priority limitations disclosed. The reference search was not exhaustive.

Metric	Criterion	Score	Manuscript-specific justification
M14	Potential impact	5.5	Provides a reproducible specialist benchmark and pedagogy for nonregular magnetic graphs, with no demonstrated broader physical impact.
M15	Gauge	9.3	Spherical holonomy, explicit vertex-gauge law, fixed cube gauge and compensated proper rotations are correctly separated from gauge-invariant spectra.
M18	Algebra	9.5	Integral characteristic polynomials, exact Gaussian-integer Gram identities, rotation multiplication and Laurent moment checks are mutually consistent.

Conditional breakthrough and interdisciplinary-unification metrics M16 and M17 are inactive because no such claims are made in the standalone note.

4 Independent mathematical verification

4.1 Graph geometry and characteristic polynomials (pp. 2–4)

The truncated octahedron has vertices the permutations of $(0, \pm 1, \pm 2)$. Its six squares have equations $x_a = \pm 2$; its eight hexagons have equations $\varepsilon \cdot x = 3$, $\varepsilon \in \{\pm 1\}^3$. Taking one graph vertex per face, the degree sequence and edge partition are

$$|V| = 14, \quad |E| = 24 \text{ square-hexagon} + 12 \text{ hexagon-hexagon} = 36, \quad d = (4^6, 6^8).$$

Every original polyhedron vertex belongs to three faces, producing 24 triangular plaquettes in the dual graph. A separate enumeration of undirected simple four-cycles returns 42. These counts must not be conflated with *induced* four-cycles, since a simple cycle can have a chord. The adjacency polynomial obtained by exact integer arithmetic is

$$\det(xI - A) = x^2(x+1)^3(x+3)(x^2-3x-12)(x^2-x-4)^3.$$

The Laplacian polynomial is

$$\det(xI - \Delta) = x(x-9)(x-7)^4(x-4)^2(x^2-9x+16)^3. \quad (1)$$

Both agree with the manuscript. The representation decomposition

$$\mathbb{R}^{14} \cong 2A_{1g} \oplus E_g \oplus 2T_{1u} \oplus T_{2g} \oplus A_{2u}$$

explains, rather than merely describes, the multiplicities. The nontrivial orthonormal Laplacian blocks are

$$\Delta_{A_{1g}} = \begin{pmatrix} 4 & -\sqrt{12} \\ -\sqrt{12} & 3 \end{pmatrix}, \quad \Delta_{T_{1u}} = \begin{pmatrix} 4 & -2 \\ -2 & 5 \end{pmatrix}.$$

Their polynomials are $x(x-7)$ and $x^2-9x+16$; the latter occurs three times. The scalar components contribute $4^{(2)}$, $7^{(3)}$, 9, exhausting all fourteen dimensions. The first block also explains why the 7-eigenspace contains an A_{1g} line in addition to T_{2g} .

4.2 Eigenspace support and the K operator (pp. 4–5)

For any eigenspace W the square-orbit support is the basis-independent average

$$\omega_s(W) = \frac{\text{tr}(P_s P_W)}{\dim W}.$$

Direct eigenspace reconstruction confirms $\omega_s(E_0) = 3/7$, $\omega_s(E_4) = 1$, $\omega_s(E_7) = 1/7$, and $\omega_s(E_9) = 0$. For the three-dimensional T_{1u} eigenspaces the values are $(1 \pm 1/\sqrt{17})/2$. The antisymmetric operator

$$K = P_s \Delta P_h - P_h \Delta P_s$$

exchanges square and hexagon components. On each normalized T_{1u} plane it is $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$, so $K^2 = -4I$ there. On the orbit-constant A_{1g} plane it has $K^2 = -12I$. It is therefore important that no global identity $K^2 = -4I$ is claimed; Draft 7 handles that restriction correctly.

4.3 Flux classification and third moment (pp. 5–7)

The 24 triangular plaquettes and Euler relation $14 - 36 + 24 = 2$ make the spherical gauge argument transparent. The oriented face boundaries span the cycle space with one dependence. Thus fixed face holonomies whose product is one determine an edge connection up to vertex gauge. Uniform holonomies require $24\Phi \in 2\pi\mathbb{Z}$ and are indexed by q modulo 24. Complex conjugation produces $q \leftrightarrow -q$ isospectrality.

For $\Delta_q = D - A_q$ the first two moments do not depend on q . In the cubic expansion, cyclicity (not commutativity) gives

$$\text{tr } \Delta_q^3 = \text{tr } D^3 + 3 \text{tr}(DA_q^2) - \text{tr } A_q^3.$$

Here $\text{tr } D^3 = 2112$, $\text{tr}(DA_q^2) = 384$, and the 24 oriented triangles give $\text{tr } A_q^3 = 144 \cos(\pi q/12)$. Consequently

$$\text{tr } \Delta_q^3 = 3264 - 144 \cos\left(\frac{\pi q}{12}\right). \quad (2)$$

The values for representatives $q = 0, \dots, 12$ are all distinct; hence any two isospectral sectors have $p \equiv \pm q \pmod{24}$. Conversely these sectors are isospectral by gauge equivalence and conjugation. The conclusion of exactly **13 distinct spectra** is therefore an analytic theorem, not an inference from the numerical plot.

4.4 Quarter and half flux (pp. 7–9)

At $q = 6$ the induced cube is flat since each cube face encloses four triangular plaquettes and $i^4 = 1$. In the explicit square–hexagon gauge (7), with C the ordinary cube adjacency and B the complex incidence block, exact matrix multiplication confirms

$$B^*B = \frac{9I - C^2}{2} = 4(P_1 + P_2), \quad BB^* = 4I_6.$$

The corresponding magnetic Laplacian is $\Delta_6 = \begin{pmatrix} 4I_6 & -B \\ -B^* & 6I_8 - C \end{pmatrix}$. For $Ch = \mu h$ in $W_1 \oplus W_2$, the normalized square partner $Bh/2$ gives the two-dimensional block

$$\begin{pmatrix} 4 & -2 \\ -2 & 6 - \mu \end{pmatrix}.$$

For $\mu = 1$ its polynomial is $x^2 - 9x + 16$; for $\mu = -1$ it is $(x - 3)(x - 8)$. Each case has three copies. The two uncoupled cube sectors give eigenvalues 3 and 9. Thus

$$\det(xI - \Delta_6) = (x - 9)(x - 8)^3(x - 3)^4(x^2 - 9x + 16)^3.$$

At $q = 12$ taking negative phase on each edge realizes every triangular holonomy -1 , so $\Delta_{12} = D + A$; exact arithmetic gives

$$\det(xI - \Delta_{12}) = (x - 8)^3(x - 5)^3(x - 4)^2(x - 3)^4(x^2 - 13x + 24).$$

The proper-rotation construction $R_s(g) = BP(g)B^*/4$ passes the required unitary, intertwining and monomial checks for all 24 rotations, as well as the multiplication checks for all 576 ordered pairs. An additional nonconstant vertex-phase transformation left the $q = 6$ eigenvalues unchanged to numerical precision; this is a gauge-covariance check, not an independent proof of the complete spherical classification.

4.5 Fourth moment, tree count and Fedorov table (pp. 6, 9–13)

Independent closed-walk counting supports the additional identity in Remark 6.3:

$$\mathrm{tr} \Delta_q^4 = 25080 - 3072 \cos \frac{\pi q}{12} + 288 \cos \frac{\pi q}{6} + 48 \cos \frac{\pi q}{3}. \quad (3)$$

At zero flux this yields 22,344, equal to the fourth power sum of the spectrum in Eq. (1). A separate exact cofactor of Δ verifies Kirchhoff’s tree count $\tau(\Gamma) = 101,154,816$. The five Fedorov spectra follow from the constructions in Section 7 and Appendix A. In particular, the elongated dodecahedron contains $x^2 - 10x + 20$ (field $\mathbb{Q}(\sqrt{5})$) while the truncated octahedron contains $x^2 - 9x + 16$ (field $\mathbb{Q}(\sqrt{17})$). This is a valid finite comparison, but there is no demonstrated general physical optimality criterion.

5 Detailed findings and correction status

F1. Final archival reproducibility — minor submission-stage requirement

Location: Section 8, pp. 10–11. The manuscript points to tag `face-graph-note-v7` and says that a Zenodo deposit of the submitted state *will* accompany submission and that its identifier will be supplied later. The statement is appropriately future tense; it should not be interpreted as a completed archive.

Required action: Before submission, establish the immutable release record containing the manuscript, all author scripts, certificates, input data, figures, requirements and runner, and insert the archive identifier. Run the declared one-command check in a clean environment against that exact release and retain the command, software versions and outcome. The PDF supplied for the present review does not by itself verify the actual tag contents or its execution.

Impact: No mathematical theorem is invalidated. This is a release closure condition and the reason the recommendation remains conditional.

F2. Historical attribution — improved; optional final refinement

Location: Acknowledgements, pp. 11–12. Draft 7 moves the detailed software correction chronology into repository records, explicitly distinguishes successive review rounds from independent referees, and states that the process is not human peer review. These changes materially improve clarity. For a submitted standalone paper, the editor may prefer a shorter acknowledgment focused on the concrete mathematical contributions with an accessible versioned provenance record. This is optional; it is not a confirmed mathematics defect.

F3. Figure 1 and evidential language — now adequately qualified

Location: Introduction p. 2; Fig. 1 p. 8; Section 8 p. 11. The graph depicts ordered eigenvalues at integer charges; the joining segments are visual guides, not analytic eigenvalue branches. Draft 7 now makes this clear in both the introduction and caption. Its definitions of proof, exact computation, and numerical tolerance are also explicit. No further correction is required for this point.

F4. Numerical release limitation — assessment boundary, not an allegation

The mathematical checks reported here were performed on independently reconstructed finite matrices. The complete version-7 author script runner was not replayed from the submitted PDF. Accordingly no statement such as “the released code passed a clean installation” is asserted in this assessment. This distinction should not be turned into a manuscript error without evidence that the actual archive is faulty.

6 Effect of findings on dependent results

Claim group	Mathematical support assessed	Effect of outstanding action
Prop. 2.1, Thm. 3.1	Geometry and exact polynomials agree	Unaffected; archive improves reproducibility
Prop. 4.1, Thm. 5.1	Support/projector and restricted K checks agree	Unaffected
Props. 6.1–6.2	Cycle rank and third-moment proof consistent	Unaffected
Thm. 6.4; Prop. 6.5	Gram, special polynomials, rotations agree	Unaffected; verify released implementation
Remark 6.3	Fourth moment and tree count agree	Unaffected
Table 1, Appendix A	Spectral constructions agree with prior and fresh checks	Unaffected

There is no evidential basis for propagating a new mathematical error through these claims. This is an explicitly bounded conclusion, not a certificate that no undiscovered error exists.

7 Author and editor recommendations

For the author

1. Complete and cite the immutable submission archive; retain an exact-version clean-run record for the stated verification entry point.
2. Confirm that the final PDF, plotted CSV and certificate all originate from that release; freeze the associated source and dependency versions.
3. Optionally shorten the acknowledgements while preserving attribution of the particular mathematical contributions and accessible provenance.
4. Preserve the present distinctions between proof, exact finite checking and numerical observations. Do not extend the claim of exact eigenvalues to uncomputed charge sectors.

For the editor

The mathematical content assessed here is sufficiently developed for a specialist referee decision conditional on ordinary release completion. The appropriate editorial question is venue fit: whether an explicit but narrowly scoped example of representation-theoretic and magnetic graph analysis meets the journal’s threshold for novelty and significance. Accepting the finite algebra does not settle historical priority or a proposed physical interpretation of the graph.

8 Overall recommendation

Accept with minor revisions, subject to venue fit and final archival reproducibility. Draft 7 improves the presentation and disclosure relative to Draft 6 without changing the supported central mathematics. The ordinary Laplacian and two special magnetic polynomials, the thirteen-sector spectral classification, the restricted inter-orbit operator and magnetic rotations, and the fourth-moment/tree-count controls form a coherent specialist contribution. No critical or major mathematical error was confirmed in the assessed scope. The remaining release obligation is actionable and does not require rewriting the mathematical proofs.